

INTRO TO SYNTHESIS

A crash course in subtractive synthesis, on the Sequential Fourm





TODAY

Club Control, Bucharest

SINTEZAUR

Bucharest's community for people who love synthesis. Your guide today: Iulian.

ZEEDO

Our retail partner, and the reason there are a dozen synths in this room.

SEQUENTIAL FOURM

Today's instrument: a 4-voice analog synth that feels pressure in every key.

SINTEZAUR × ZEEDO · POWERED BY SEQUENTIAL

WHERE ON FOURM

TRANSPOSE DOWN + PROGRAM

Hold TRANSPOSE DOWN (left of the keys) and press PROGRAM: the synth resets to the Basic Program.



SIGNAL FLOW

How sound travels through a subtractive synthesizer



WHAT'S IN A SOUND

A note is a fundamental plus a stack of fainter harmonics. Their mix is the timbre.

BAR HEIGHT = HOW LOUD · LEFT TO RIGHT = HIGHER IN PITCH



SAW

EVERY HARMONIC

bright, buzzy



SQUARE

ODD HARMONICS ONLY

hollow, woody



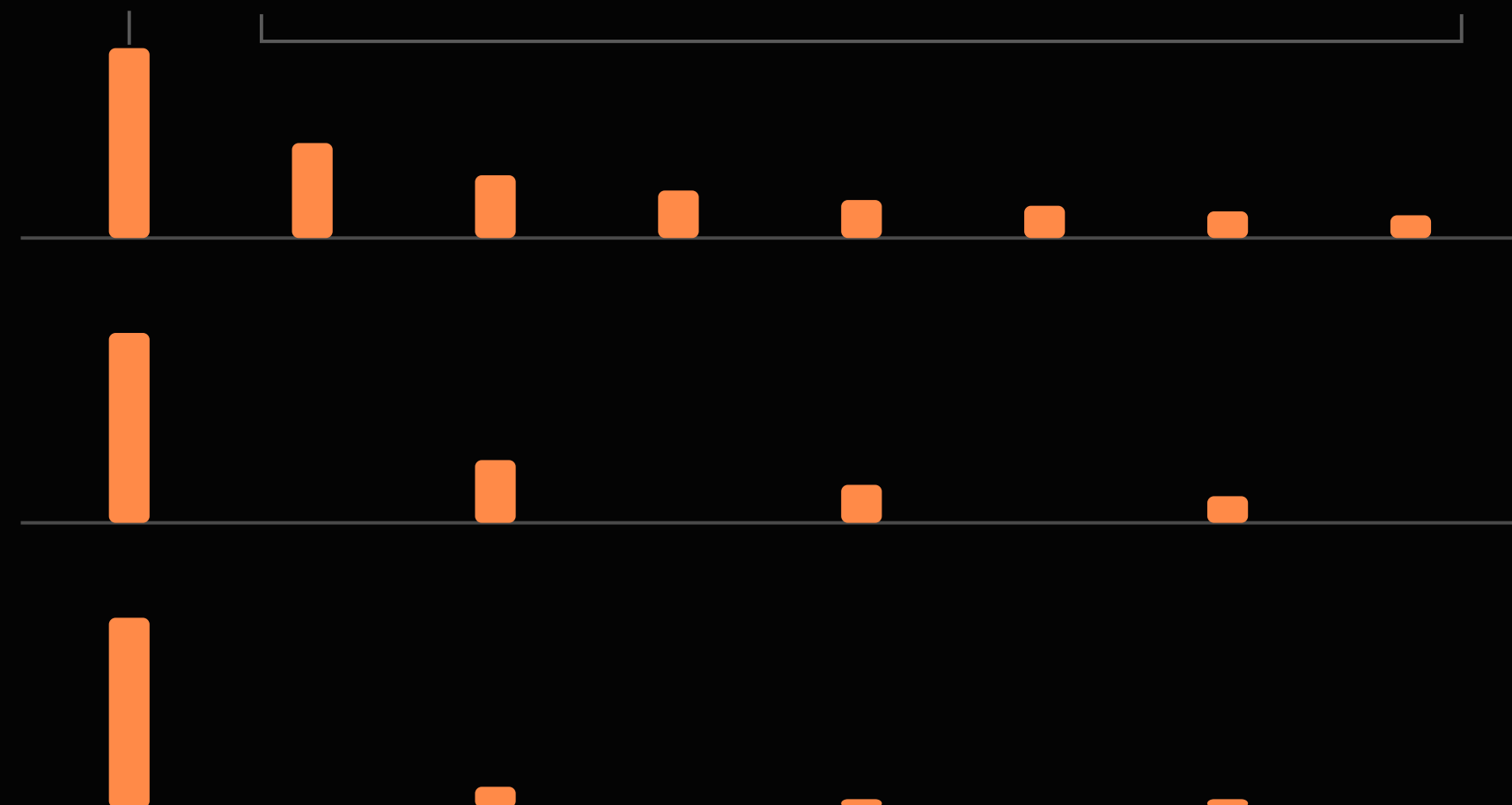
TRIANGLE

ODD, FADING FAST

soft, nearly a sine

FUNDAMENTAL

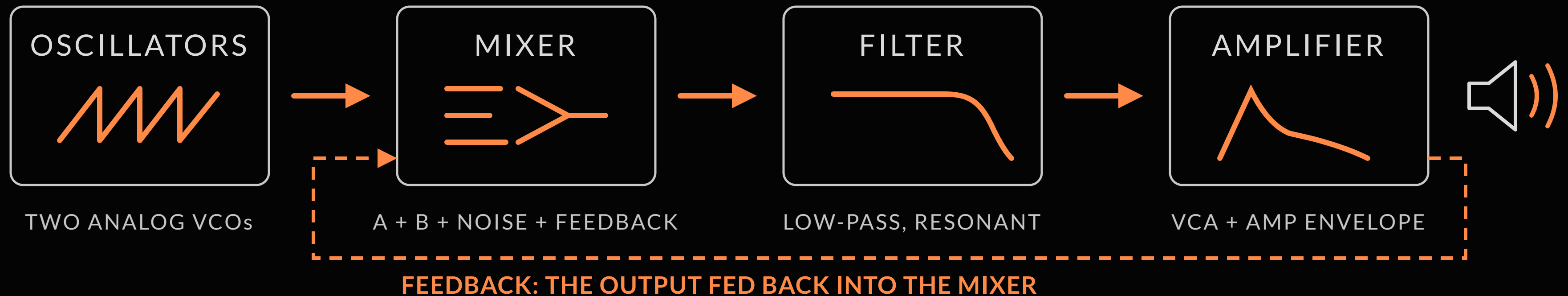
HARMONICS



SQUARE IS PULSE AT 50%. OFF 50%, EVEN HARMONICS APPEAR: THAT'S WHAT PWM MOVES.

THE VOICE PATH

Oscillators, mixer, filter, amplifier: one road for every sound



WHERE ON FOURM

The signal path, left to right across the panel



THE MIXER

Every voice starts here: four sources feed the filter

OSC A



SAW + PULSE
THE BRIGHT BACKBONE

OSC B



TRI + SAW + PULSE
WEIGHT, DETUNE, LO MODES

NOISE



WHITE / PINK / VIOLET
AIR, PERCUSSION, TEXTURE

FEEDBACK



OUTPUT LOOP
SOFT-CLIP SATURATION

TURN UP AT LEAST ONE SOURCE, OR LET RESONANCE SELF-OSCILLATE

KEY TERMS

New words from this module

SIGNAL FLOW

The road a sound takes: oscillators → mixer → filter → amplifier → output.

VCO · OSCILLATOR

Voltage-controlled oscillator. Generates the raw waveform. Fourm has two.

MIXER

Sets the level of each source: Osc A, Osc B, noise, feedback.

HARMONICS

Whole-number multiples of the fundamental, stacked above it and usually fainter; their mix is timbre.

NOISE

Unpitched sound, in white, pink, or violet flavors.

FEEDBACK

The output fed back into the mixer, soft-clipped: warmth, then growl.

VCF · VCA

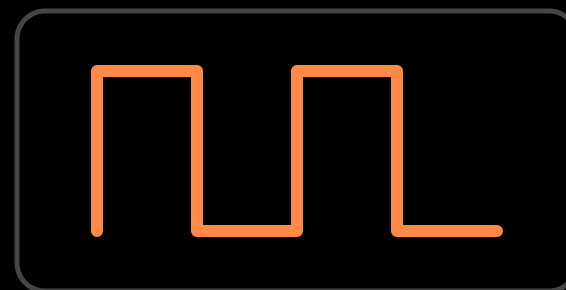
Voltage-controlled filter and amplifier: the tone and loudness stages.

OSCILLATORS

Where every sound is born



SAW



PULSE



TRI

THE WAVESHAPES

Two oscillators, five buttons, endless timbres



SAW: ALL HARMONICS · PULSE: ODD ONLY WHEN SQUARE



ADDS WEIGHT, DETUNE, LO MODES

LIGHT SEVERAL SHAPE BUTTONS: THE WAVES ADD UP

OCTAVES & TUNING

Detune B against A and the sound starts to breathe.

-2

-1

0

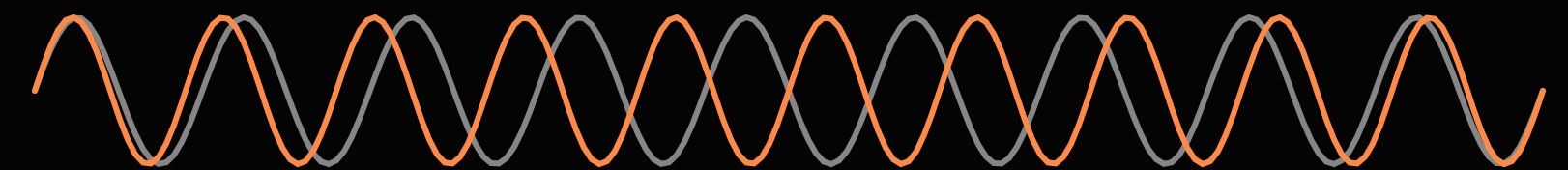
+1

+2

FIVE OCTAVE POSITIONS ON EACH OSCILLATOR

OSC B ONLY: LO1 · LO2, BELOW THE OCTAVES —
SUB-AUDIO, IT BECOMES A MODULATOR (LEVEL 2)

FREQUENCY · FINE TUNE · ± 7 SEMITONES



A AND B, A FEW CENTS APART



TOGETHER THEY BEAT: THE SOUND BREATHES

TOO FAR AND YOU HEAR TWO NOTES: BRING IT BACK UNTIL IT JUST SHIMMERS

WHERE ON FOURM

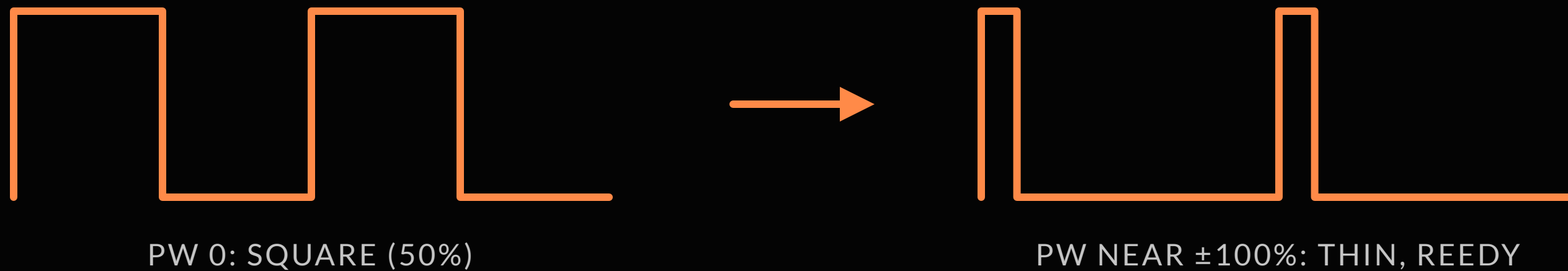
OSC B › OCTAVE · FREQUENCY

OCTAVE is the encoder. Turn FREQUENCY slowly off centre: slow beats, chorus, two notes, then back.



PULSE WIDTH & PWM

No PW knob: hold PULSE, turn SELECT. 0 = square. Toward ± 100 : thin, then silence.



AT FULL $\pm 100\%$ THE PULSE NARROWS TO SILENCE

ROUTE LFO OR FILT ENV TO PW: THAT MOVEMENT IS PWM

WHERE ON FOURM

OSC A › PULSE|PW + SELECT

Hold PULSE|PW and turn SELECT: zero is the square, towards the edge the pulse thins to a thread.



THE SUB TRICK

Osc B, one octave down, triangle

OSC A: SAW · OCTAVE 0



OSC B: TRI · OCTAVE -1



TOGETHER



THE TRIANGLE ADDS A CLEAN OCTAVE BELOW: WEIGHT WITHOUT MUD

NOISE

Three colors of texture: white, pink, violet

WHITE



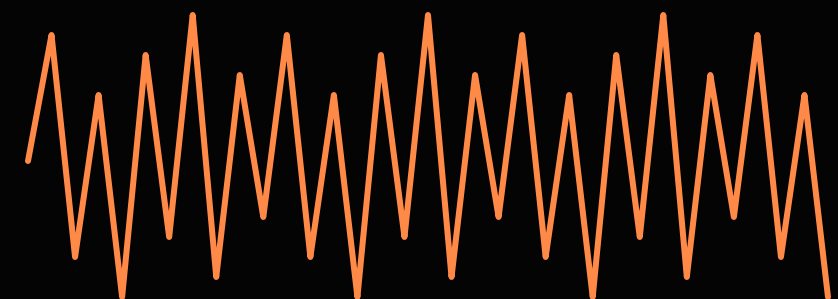
FLAT: HISS, SNARE

PINK



WARM: WIND, WAVES

VIOLET



BRIGHT: AIR, SIZZLE

NOISE COMES IN AT FULL LEVEL: HOLD NOISE AND TURN SELECT DOWN UNTIL IT SITS UNDER THE SOUND

THE COLOUR, WHITE, PINK, VIOLET, IS CHOSEN IN THE PARAM MENU

KEY TERMS

New words from this module

WAVESHAPE

The geometric shape of one cycle: it sets the raw timbre.

PWM

Pulse width modulation: moving PW with the LFO or an envelope.

LO1 / LO2

Osc B's low modes: LFO-range pitch, key-tracked (LO1) or free (LO2).

PULSE WIDTH (PW)

Duty cycle of the pulse wave, -100% to +100%. 0 = square.

SYNC

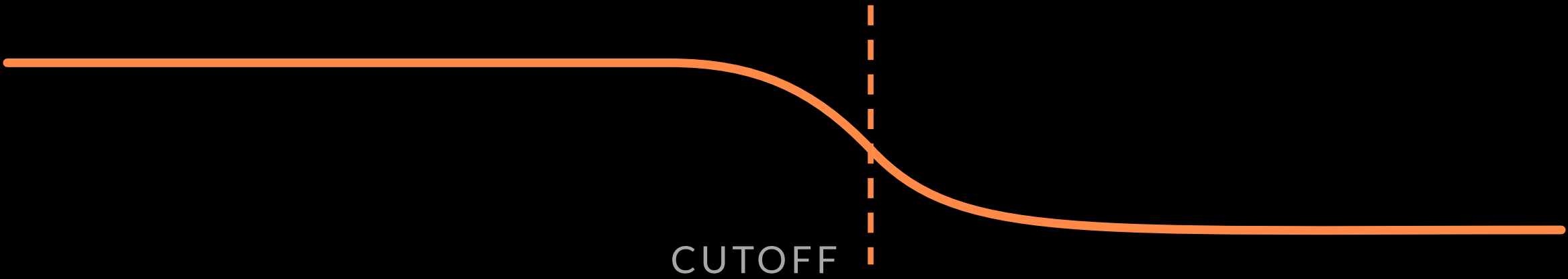
Locks Oscillator A's cycle to Oscillator B for aggressive timbres.

HARMONICS

Saw has them all; square and triangle only the odd ones, the triangle's fade much faster.

FILTER

Shaping your tone



FILTER

ENV AMT

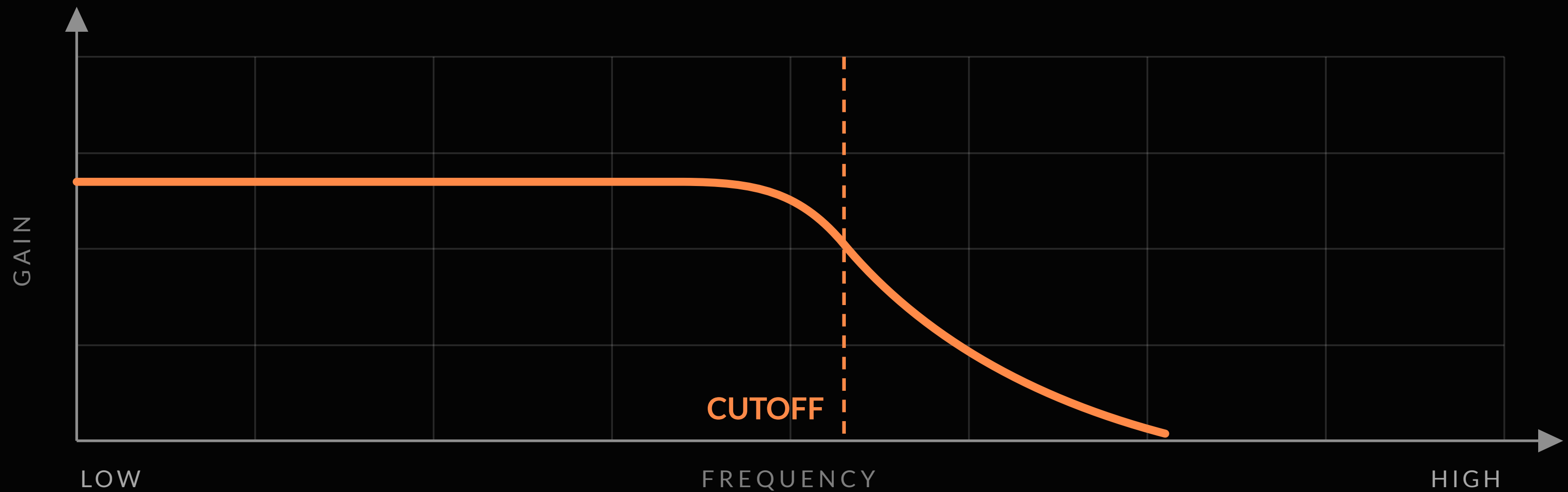
KEY TRACK AMOUNT

CUTOFF

RESONANCE

LOW-PASS

Below the cutoff passes, above it fades. Close it: darker. Open it: brighter.



WHERE ON FOURM

FILTER › CUTOFF

Close CUTOFF slowly, then open it again: the harmonics leave and come back one by one.



RESONANCE

A boost right at the cutoff. Turn it up and the filter sings; at maximum it becomes a sound source of its own.



WHERE ON FOURM

FILTER › RESONANCE

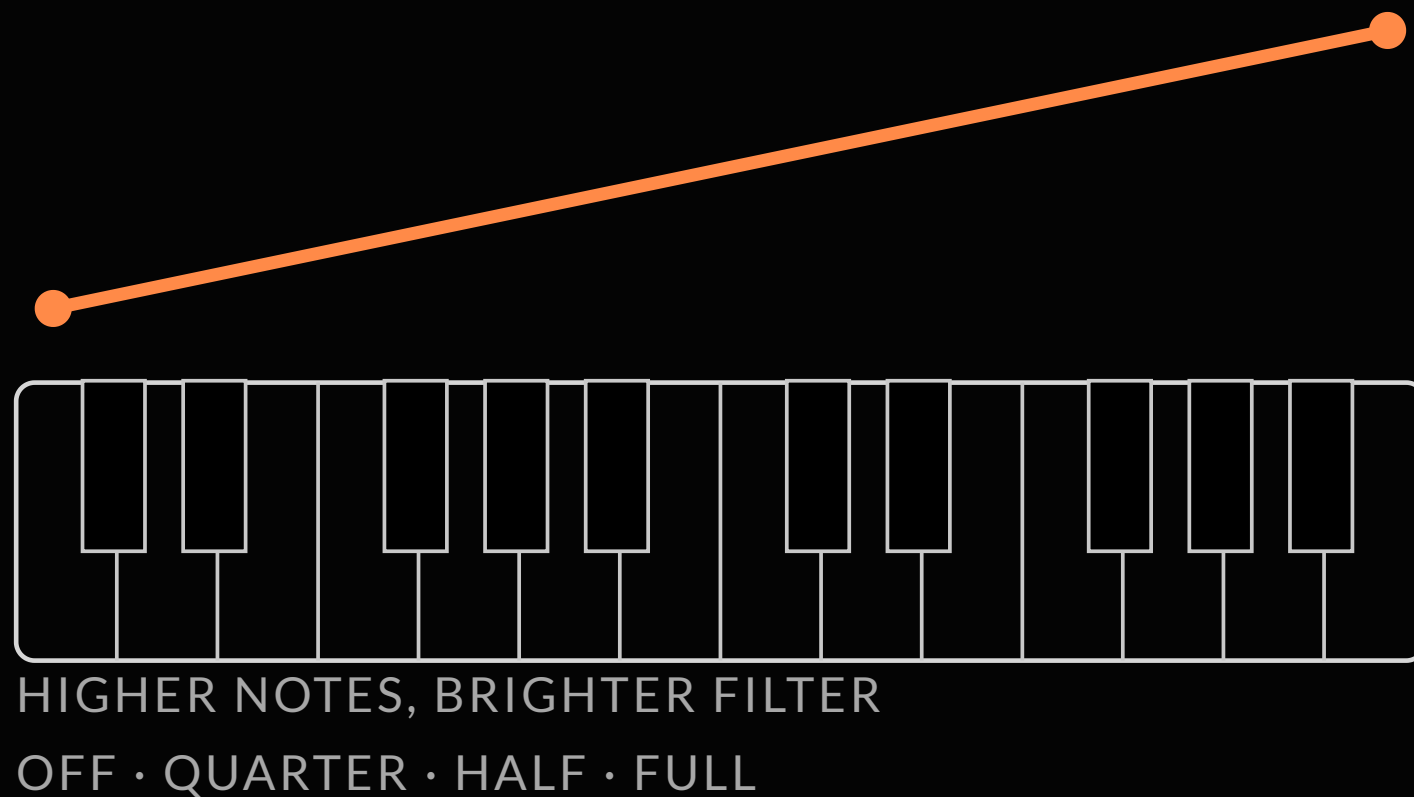
RESONANCE up while you sweep CUTOFF: a peak that whistles along; all the way up the filter self-oscillates: it sounds on its own.



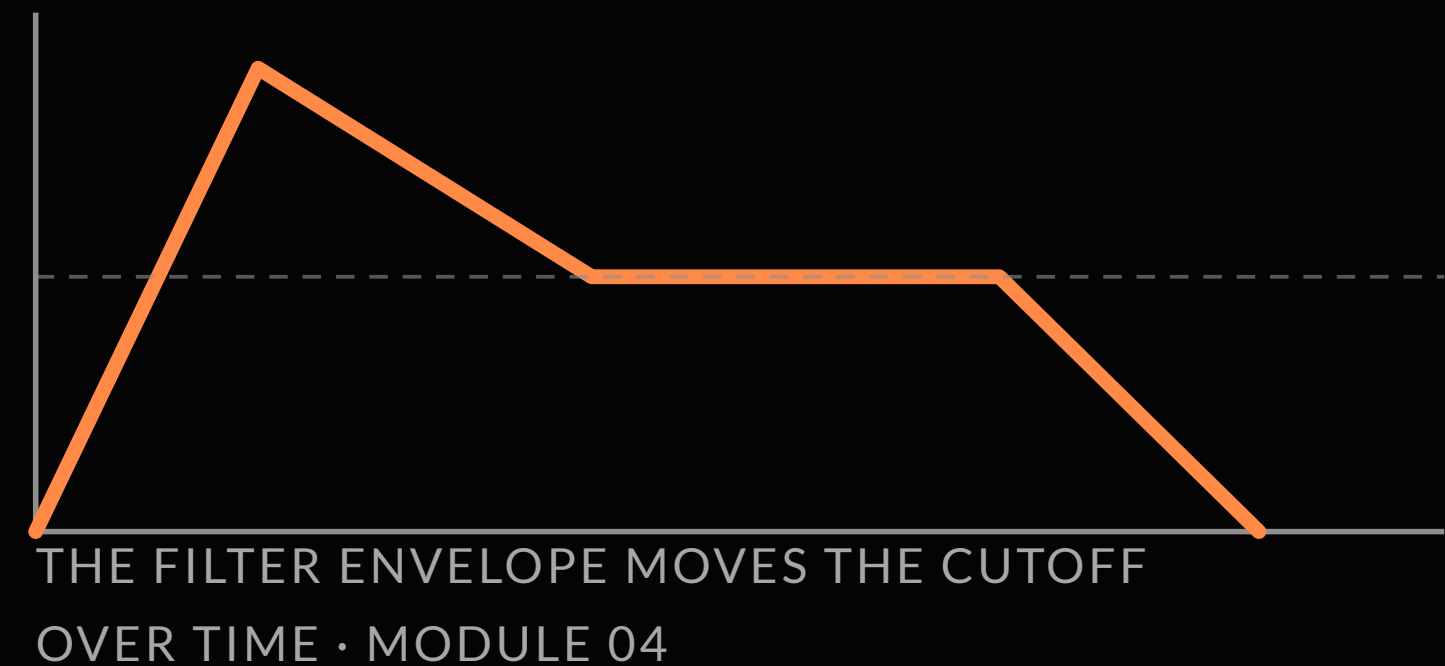
KEY TRACK & ENV AMOUNT

The keyboard opens the filter; the envelope moves it in time.

KEY TRACK



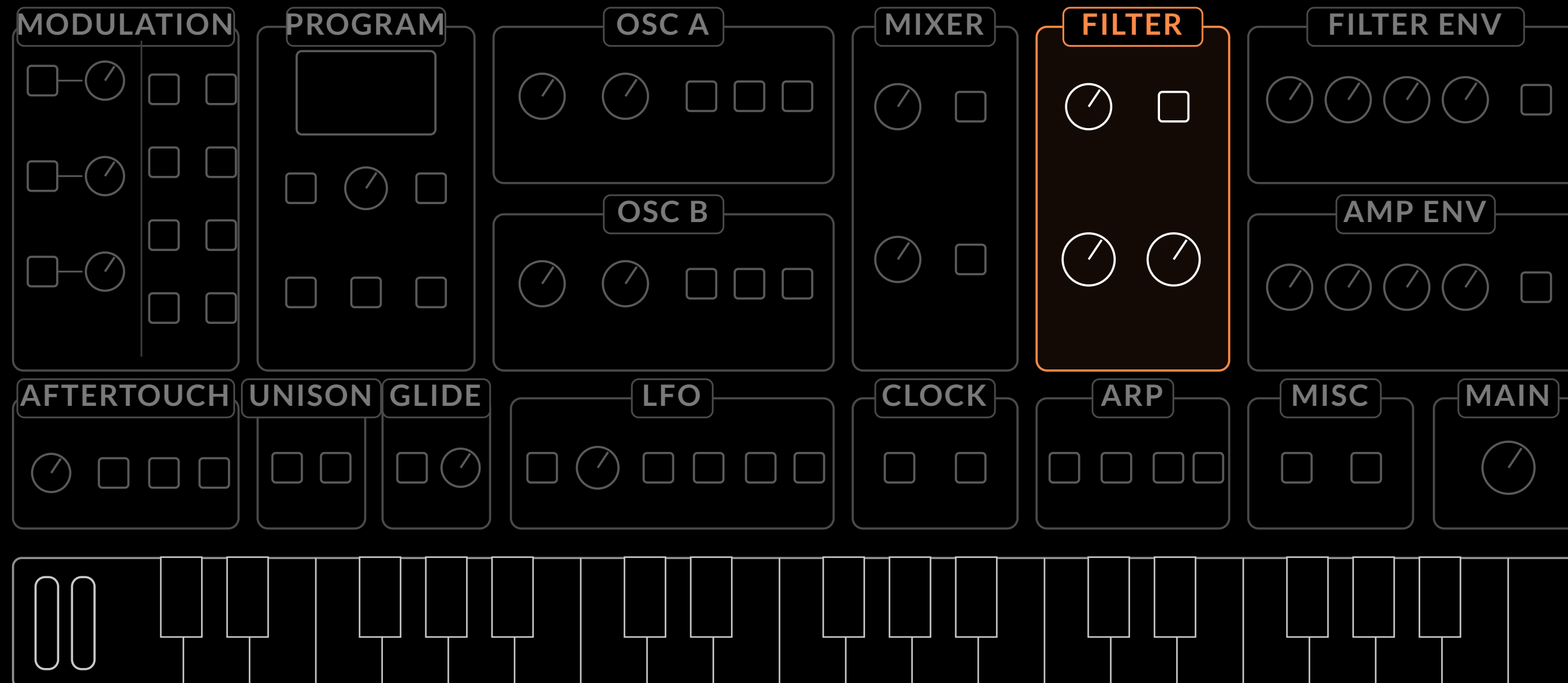
ENV AMT



WHERE ON FOURM

FILTER › KEY TRACK · ENV AMT

KEY TRACK on: high notes open the filter more. ENV AMT: how far the envelope pushes the cutoff.



KEY TERMS

New words from this module

CUTOFF

The hinge: below passes, above fades.

SELF-OSCILLATION

At maximum, the filter becomes a sine source.

ENV AMOUNT

How much the envelope moves the cutoff.

RESONANCE

A boost right at the cutoff.

KEY TRACK

Higher notes, brighter filter.

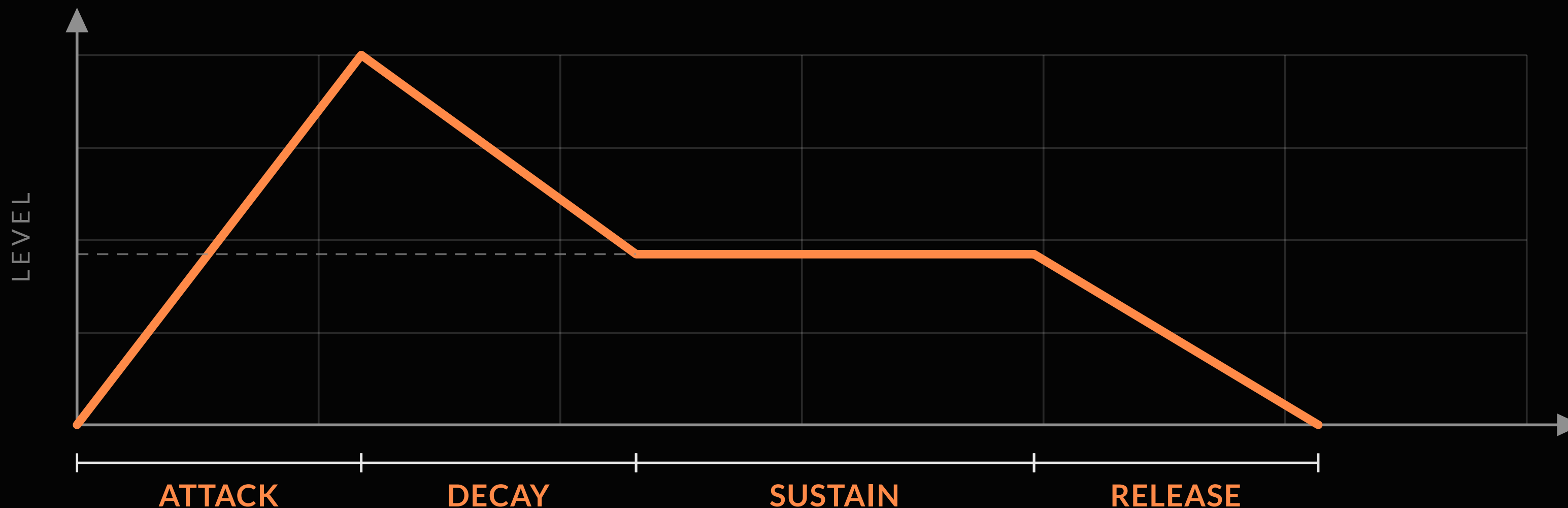
ENVELOPES

Controlling your sound over time



ADSR

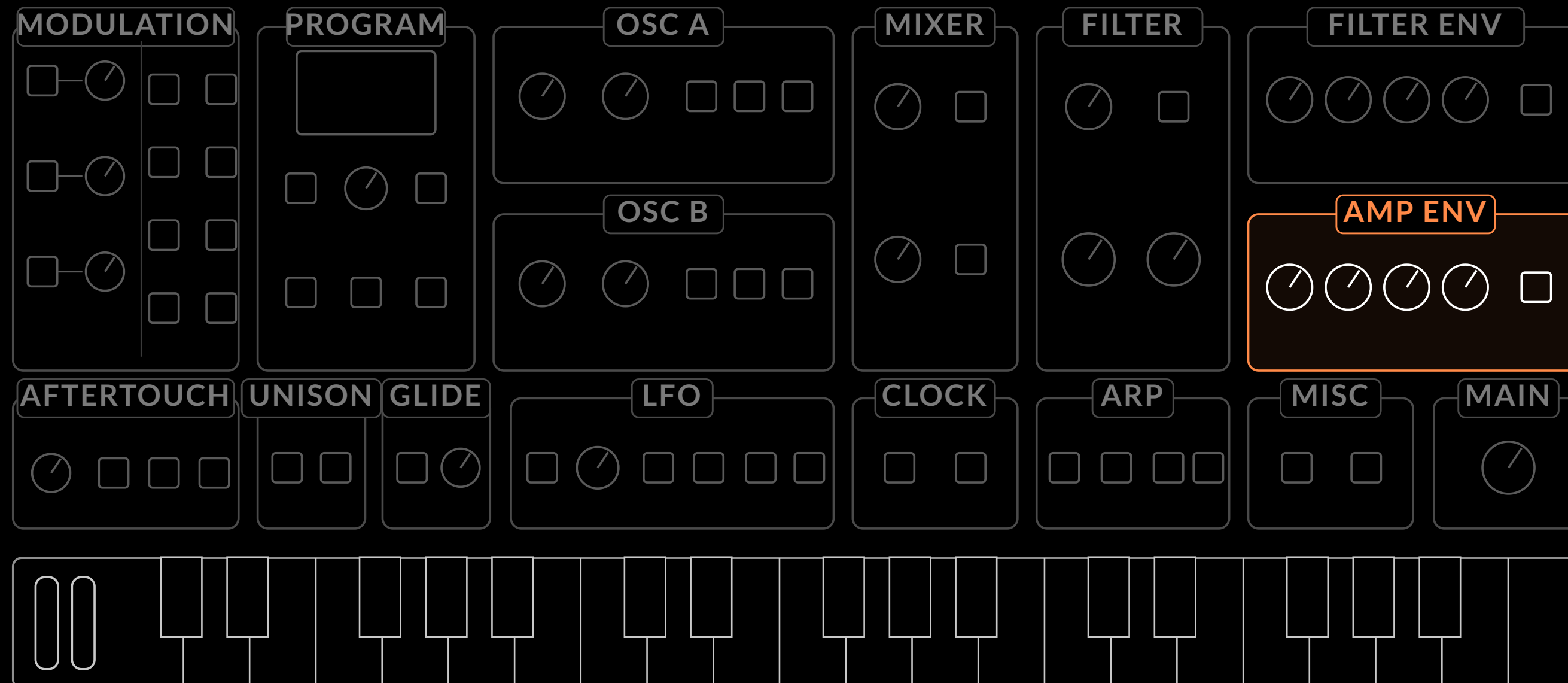
A shape that happens every time you press a key. Sustain is a level, not a time.



WHERE ON FOURM

AMP ENVELOPE › ATTACK · DECAY · SUSTAIN · RELEASE

Four knobs, one shape. Play a note, turn ATTACK up, play it again; then let go and hear RELEASE. The note learns to breathe.



TWO ENVELOPES

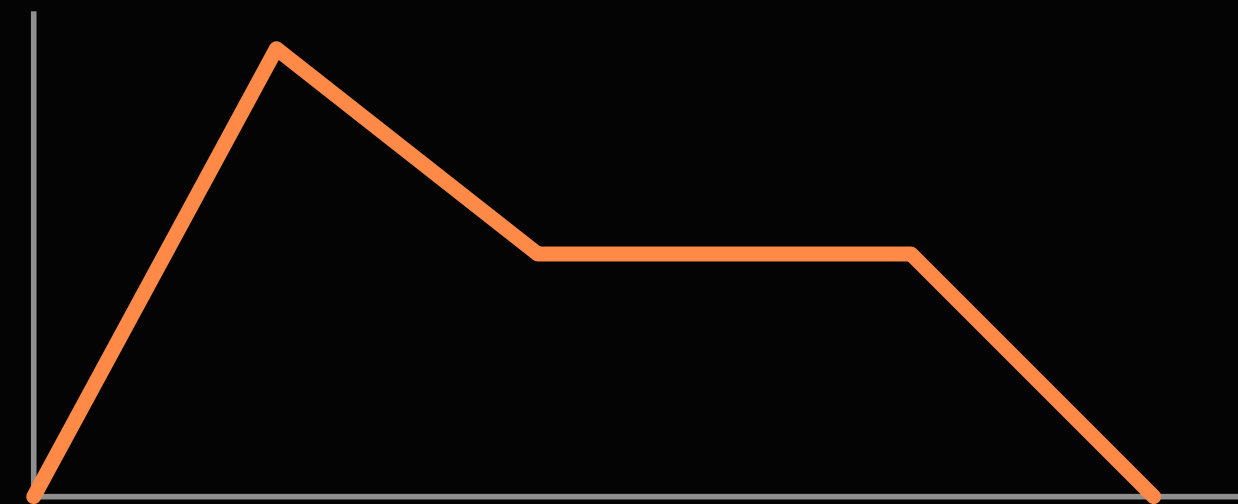
Same shape, two jobs: how loud, how bright.

AMP ENVELOPE



LOUDNESS
HOW LOUD

FILTER ENVELOPE

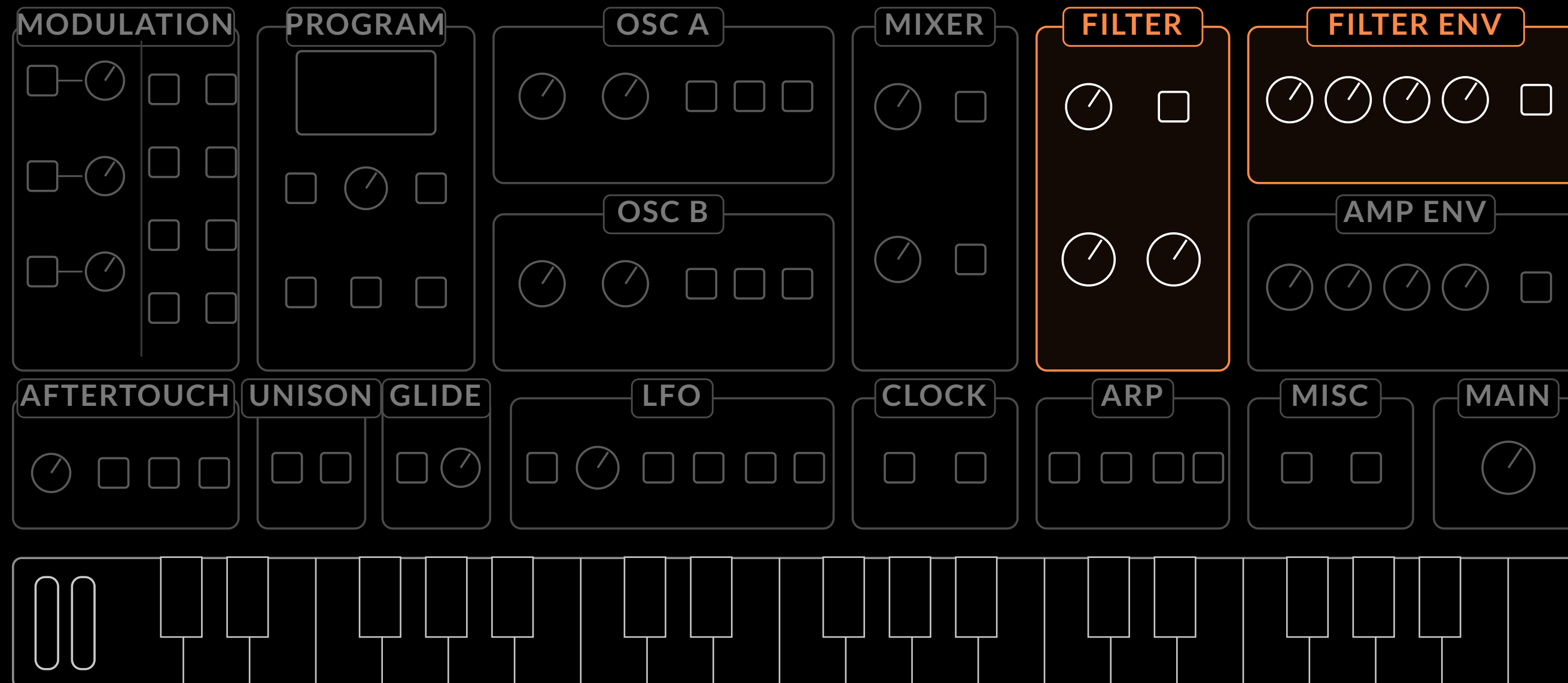


CUTOFF · SCALED BY ENV AMT
HOW BRIGHT

WHERE ON FOURM

FILTER ENV › DECAY + FILTER › ENV AMT

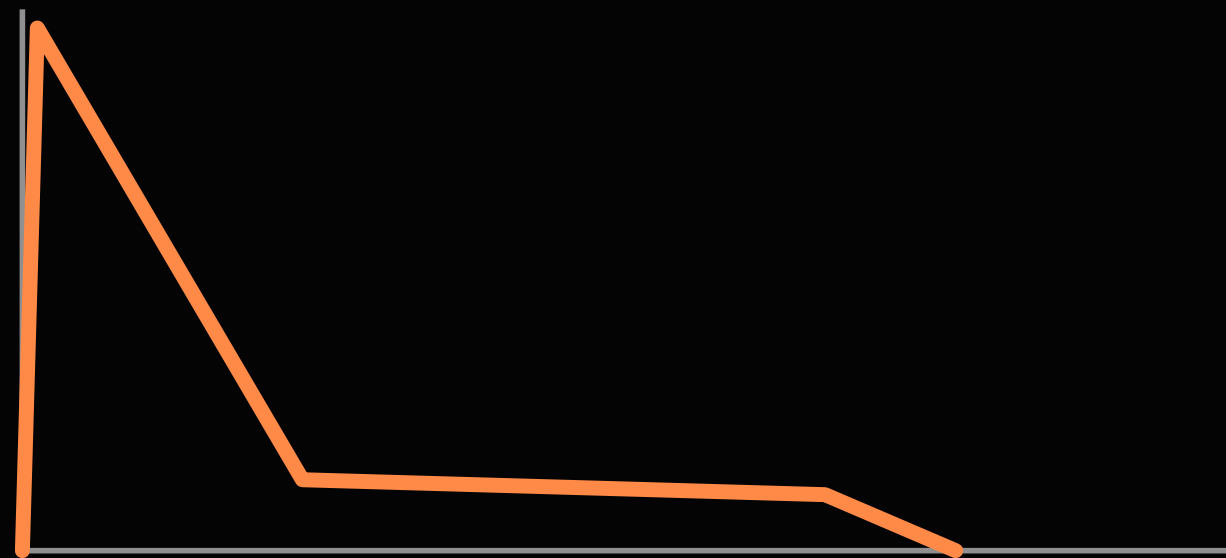
ENV AMT (in the filter) sends the envelope to the cutoff; DECAY sets how fast the brightness falls away.



BASS vs PAD

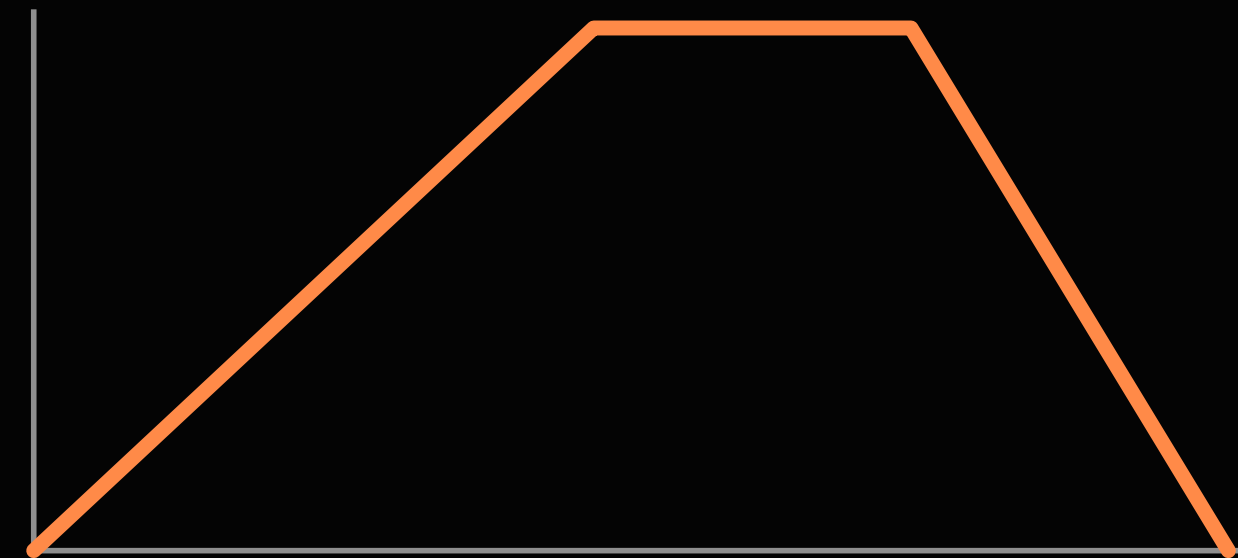
The same oscillators, two instruments.

BASS



INSTANT ATTACK · SHORT DECAY
LOW SUSTAIN

PAD



SLOW ATTACK · NO DECAY
FULL SUSTAIN · LONG RELEASE

KEY TERMS

New words from this module

ATTACK

Time from silence to the peak.

SUSTAIN

A level, not a time.

ENV AMOUNT

How much the envelope moves the cutoff.

DECAY

Time from the peak to the sustain level.

RELEASE

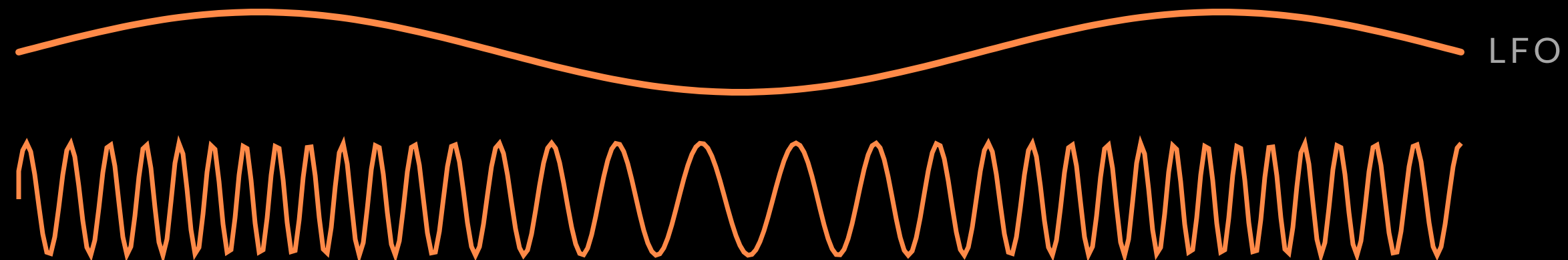
Time to fade after the key is let go.

VELOCITY

How hard you strike scales the envelope.

LFO & MODULATION

Adding movement



THE LFO

An oscillator too slow to hear. You hear what it moves.

LFO

→ PITCH **VIBRATO**



→ CUTOFF **WAH**



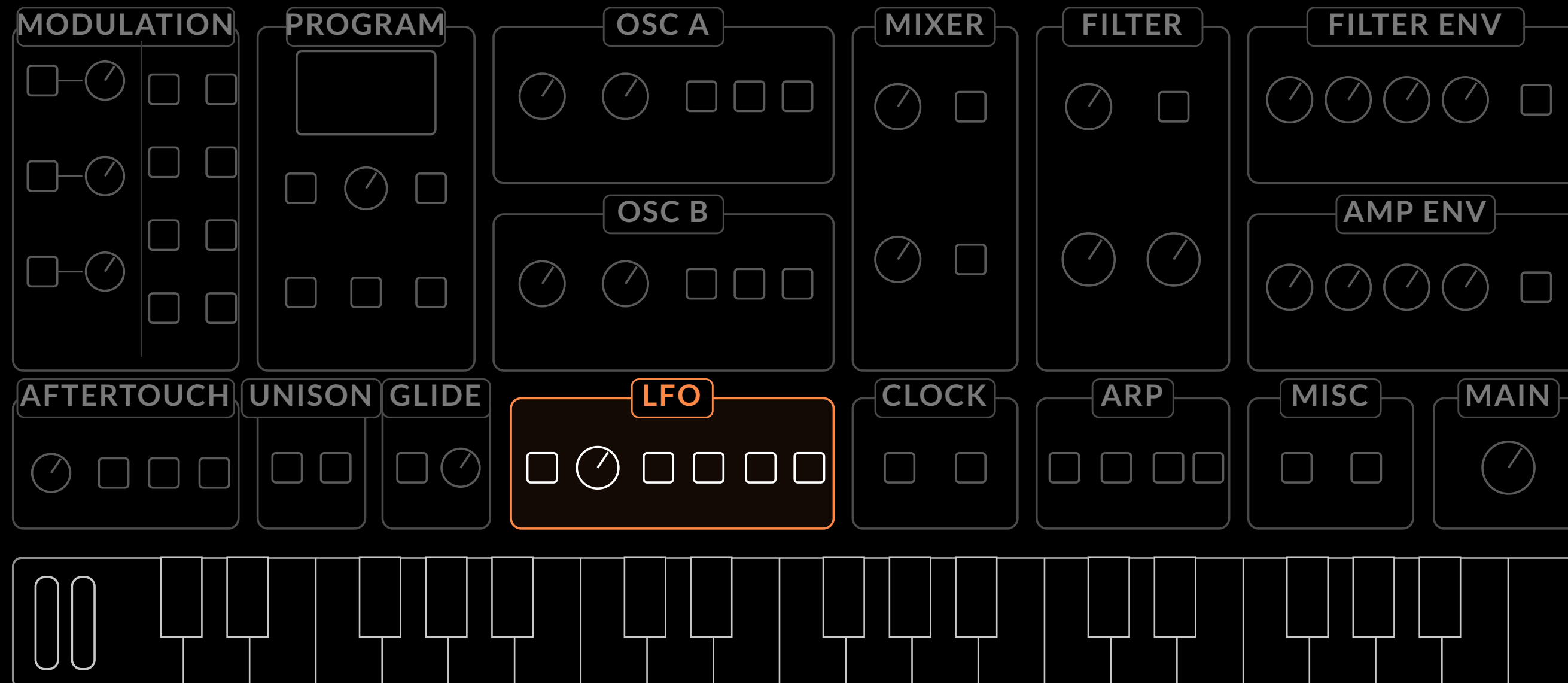
→ AMP **TREMOLO**



WHERE ON FOURM

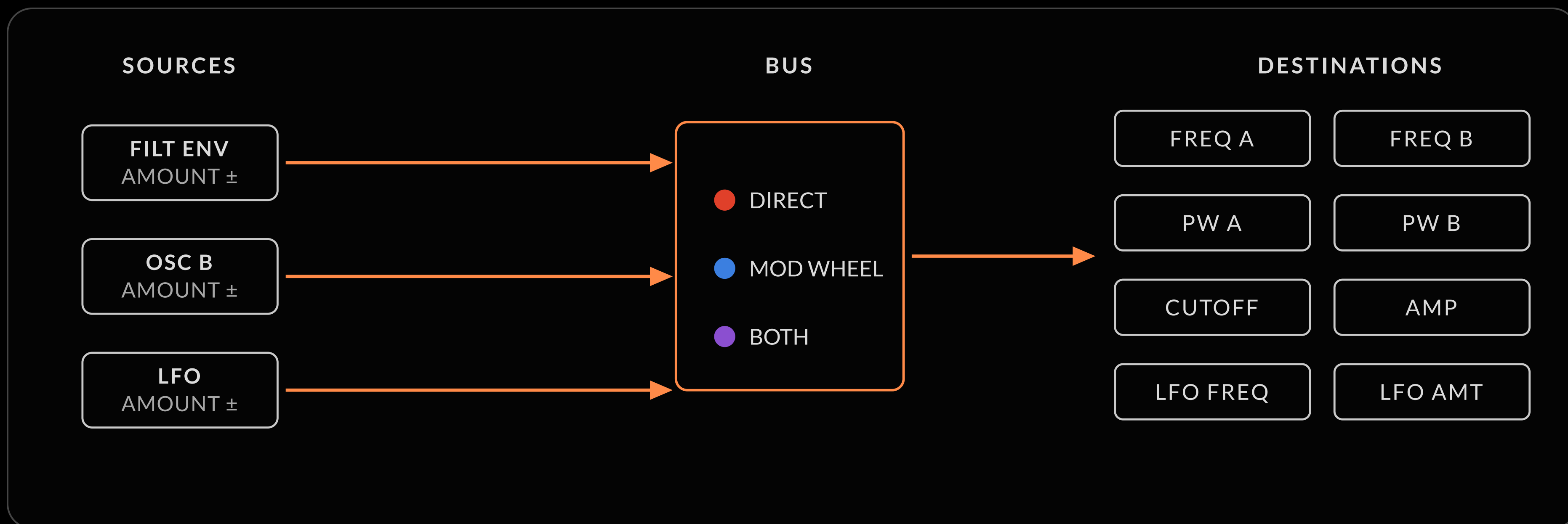
LOW FREQUENCY OSCILLATOR

Mod wheel up for vibrato: the Basic Program already wires it. FREQUENCY sets the speed, the shape buttons the motion; how far it goes is the LFO's AMOUNT knob in the MODULATION section, next slide.



THE MODULATION SECTION

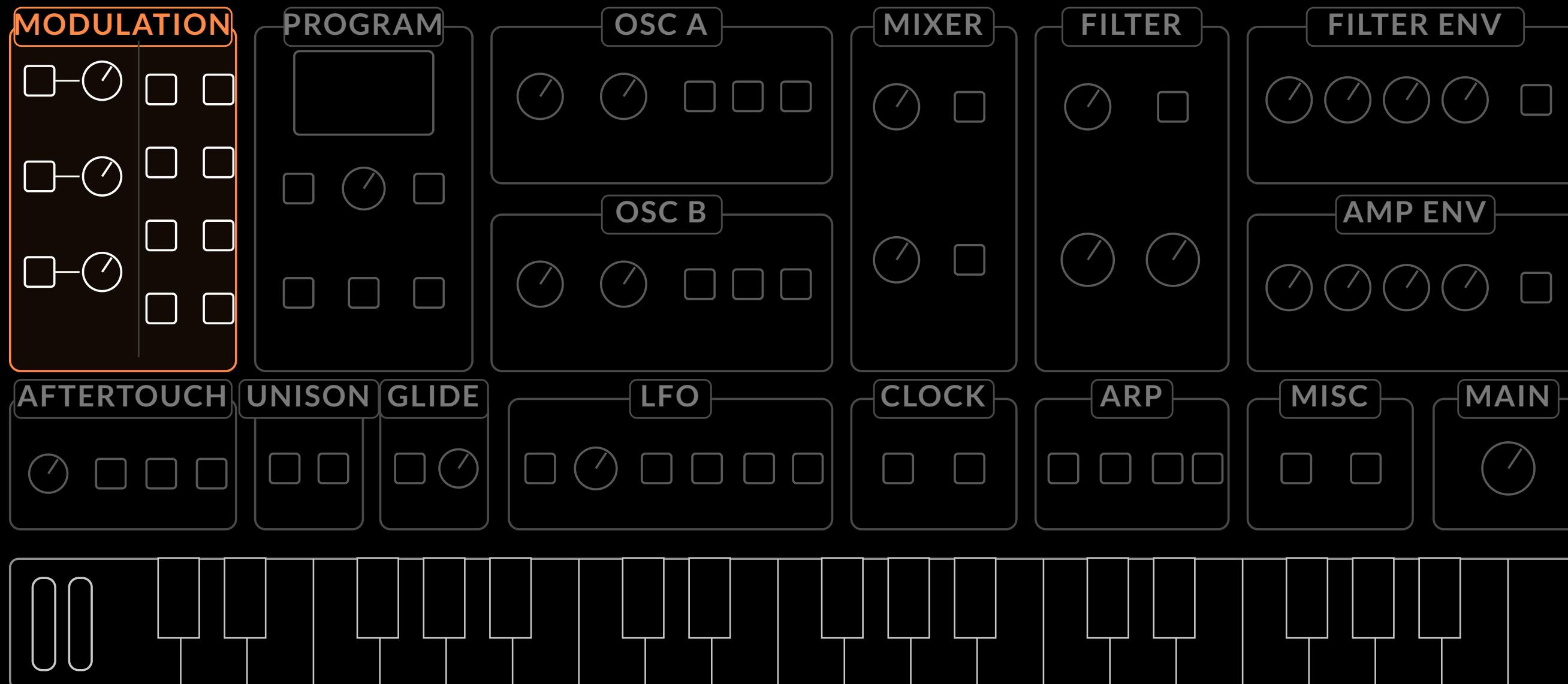
Route → destination → amount. On the wheel bus, the mod wheel is the volume of the modulation.



WHERE ON FOURM

MODULATION › SOURCE · DESTINATION · AMOUNT

Three sources. Press a ROUTE, then a destination, then AMOUNT: zero at 12 o'clock.
 · red = the direct bus, always on · blue = the mod-wheel bus · purple = both



KEY TERMS

New words from this module

LFO

An oscillator too slow to hear.

VIBRATO

LFO to pitch.

PWM

Pulse width moved by a modulator, usually the LFO.

MOD WHEEL

The volume of the modulation.

RATE • FREQUENCY

Its speed (the FREQUENCY knob on Fourm); SYNC locks it to the clock.

TREMOLO

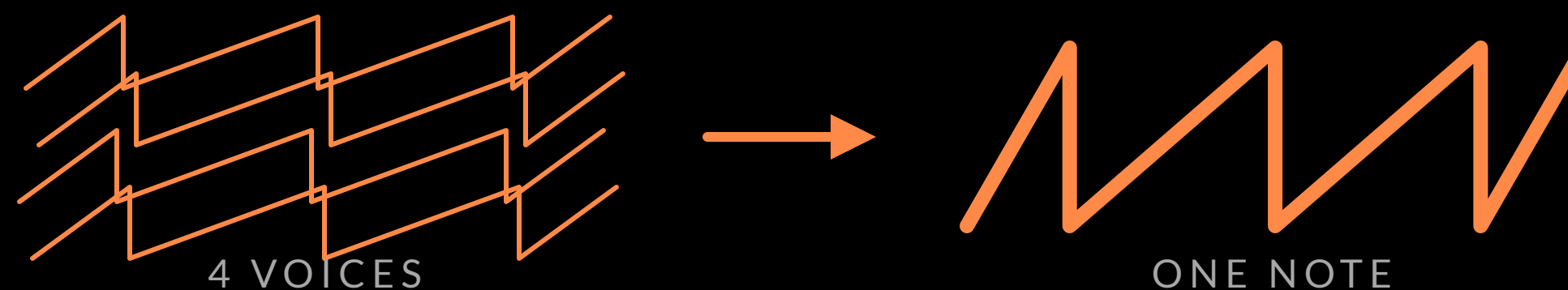
LFO to amplitude.

MODULATION BUS

Direct (red) or mod wheel (blue).

VOICES & UNISON

Four voices, one note, and the loop that thickens everything

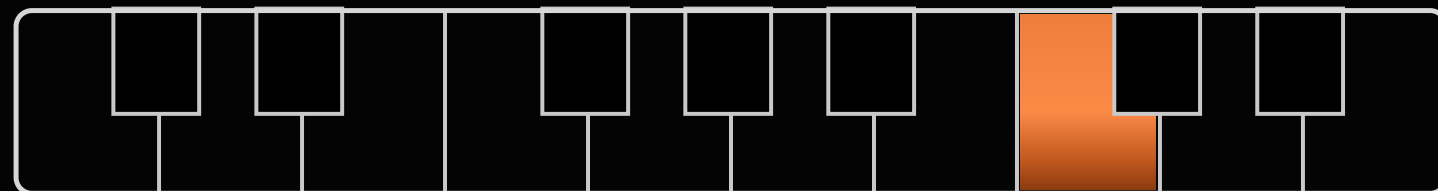


POLYPHONY

Fourm has four complete voices. Play a fifth note and the oldest one is stolen.

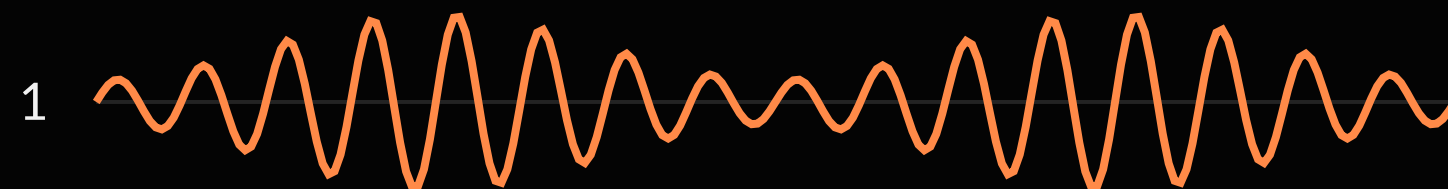
MONOPHONIC · ONE VOICE

WHAT YOU PLAY



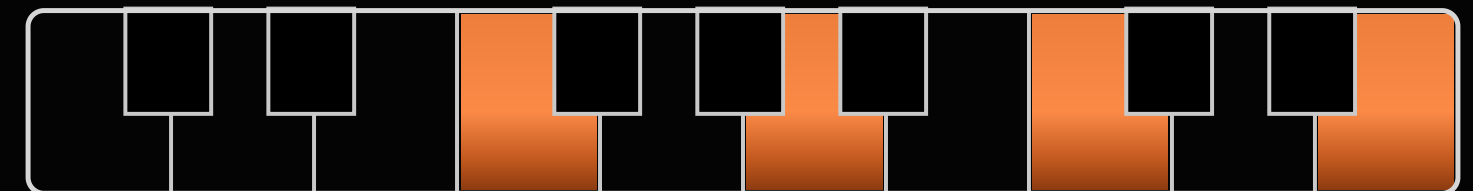
↓ LAST NOTE

WHAT YOU HEAR



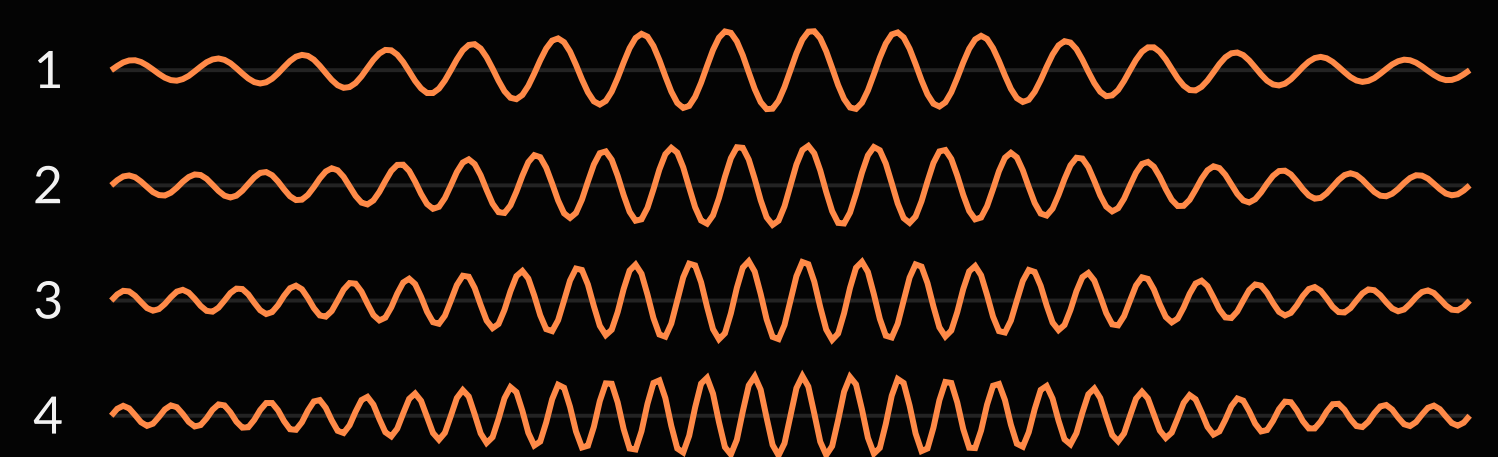
4-VOICE POLYPHONIC

WHAT YOU PLAY



↓ ↓ ↓ ↓

WHAT YOU HEAR



UNISON

Four voices on one note



1-4 VOICES, SLIGHTLY DETUNED



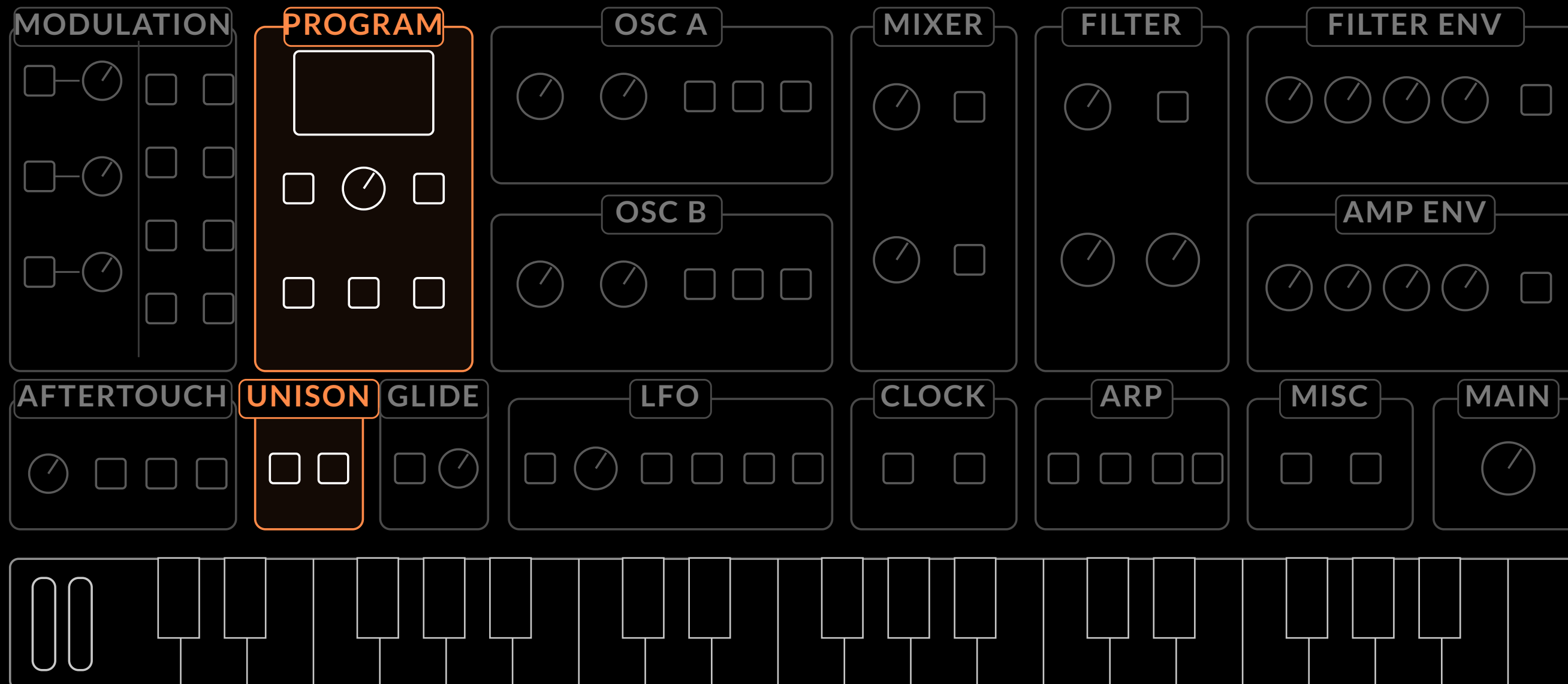
ONE HUGE MONOPHONIC NOTE

HOLD UNISON + SELECT FOR VOICES • PARAM 10: UNISON DETUNE 0-127

WHERE ON FOURM

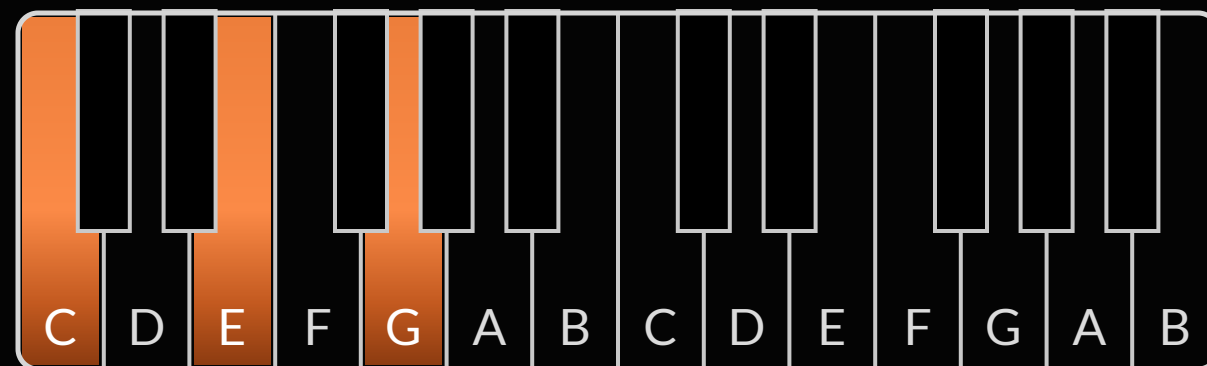
UNISON + SELECT · DISPLAY: PARAM 10

UNISON on; hold it and turn SELECT for the voice count. The detune between them lives at Param 10 on the display.

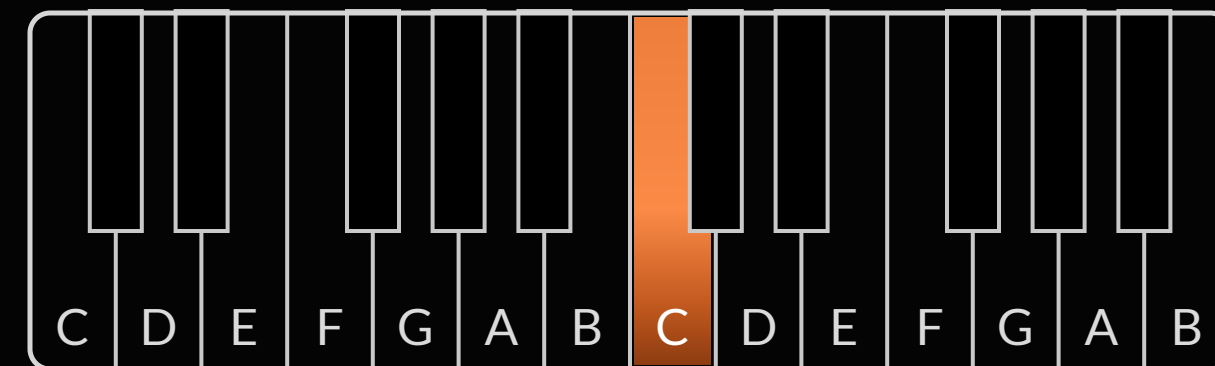


CHORD MEMORY

One key plays the whole chord



1 · HOLD A CHORD, PRESS UNISON



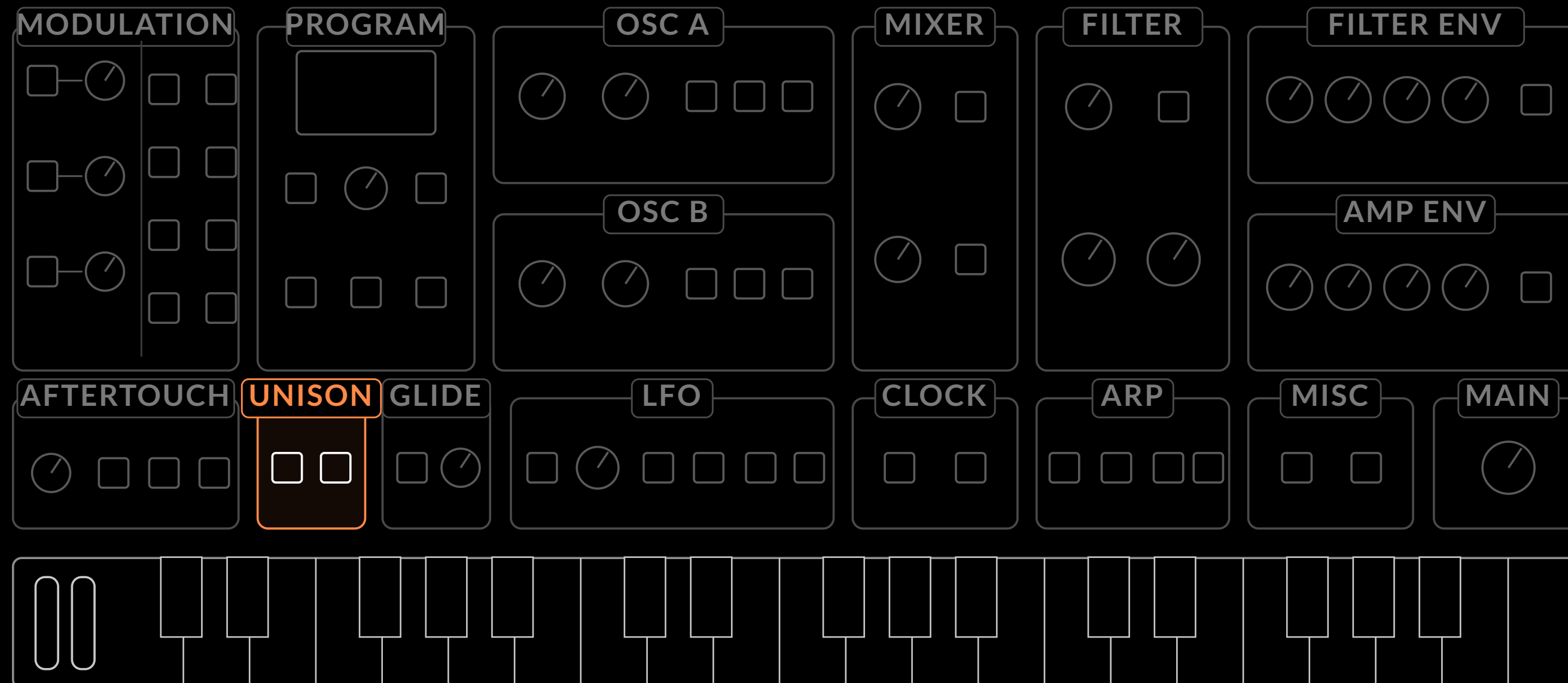
2 · ANY SINGLE KEY REPLAYS IT, TRANSPOSED

SAVED WITH THE PROGRAM: GREAT FOR CHORD STABS

WHERE ON FOURM

UNISON › CHORD MEMORY

With Chord selected in the UNISON menu, hold a chord and press UNISON: from now on any single key plays the whole chord, transposed. Full steps: handbook, module 06.



FEEDBACK

The output fed back into the mixer, soft-clipped: warmth, then growl.

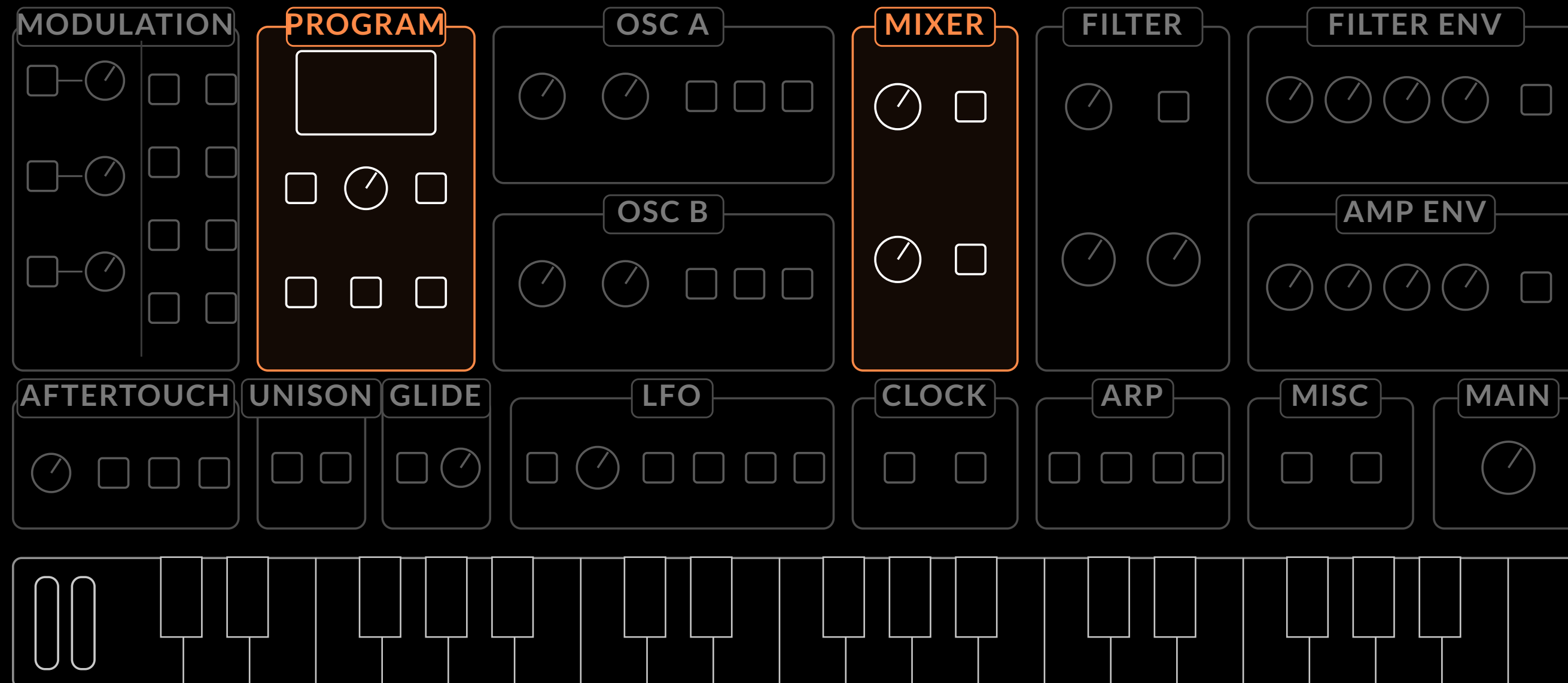


DEFAULT LEVEL 100 • HOLD FEEDBACK + TURN SELECT • THICKEN, DRIVE, CHAOS

WHERE ON FOURM

MIXER › FEEDBACK + SELECT

FEEDBACK on; hold it and turn SELECT for the amount: warmth first, then growl.



KEY TERMS

New words from this module

UNISON

All chosen voices play the same note. Monophonic, massive.

CHORD MEMORY

Unison stores a chord; single keys replay it transposed.

FEEDBACK

The output fed back into the mixer, soft-clipped.

POLYPHONY • VOICE

Four complete voices; the fifth note steals the oldest.

UNISON DETUNE

0–127 of pitch offset between stacked voices: a thickening that sounds like a chorus.

HOLD

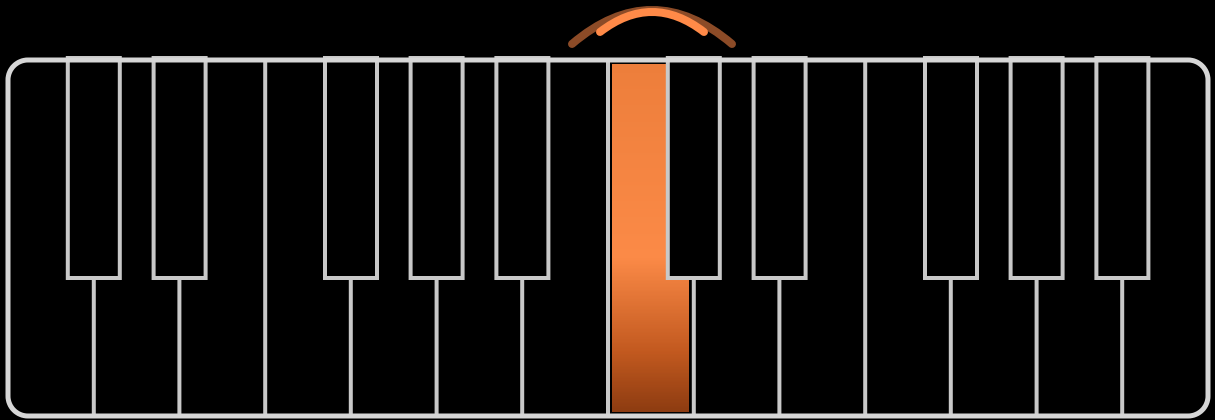
Sustains up to four notes after your hands leave, at the amp envelope's sustain level; also latches the arpeggiator.

SOFT-CLIPPING

Gentle rounding of peaks: saturation instead of harsh distortion.

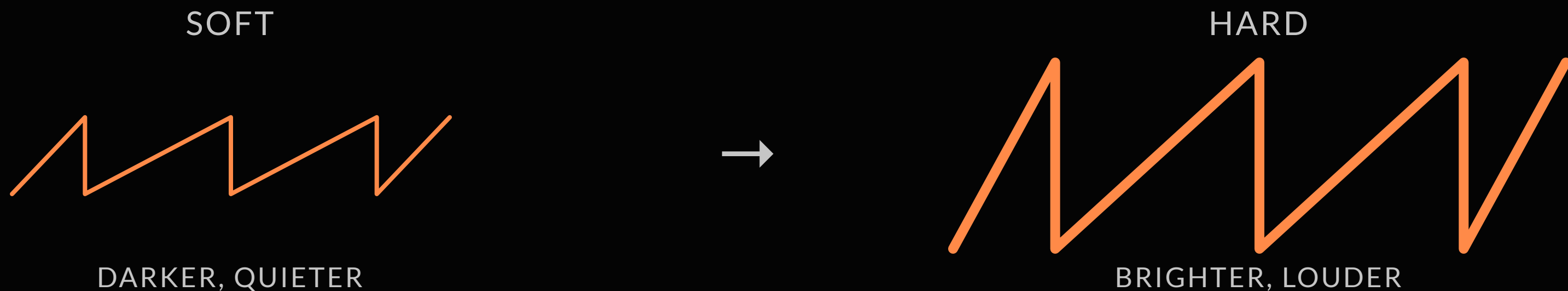
EXPRESSION

Play it, don't just press it



VELOCITY

How fast the key goes down. Scales the amp and filter envelopes; five response curves in Global.



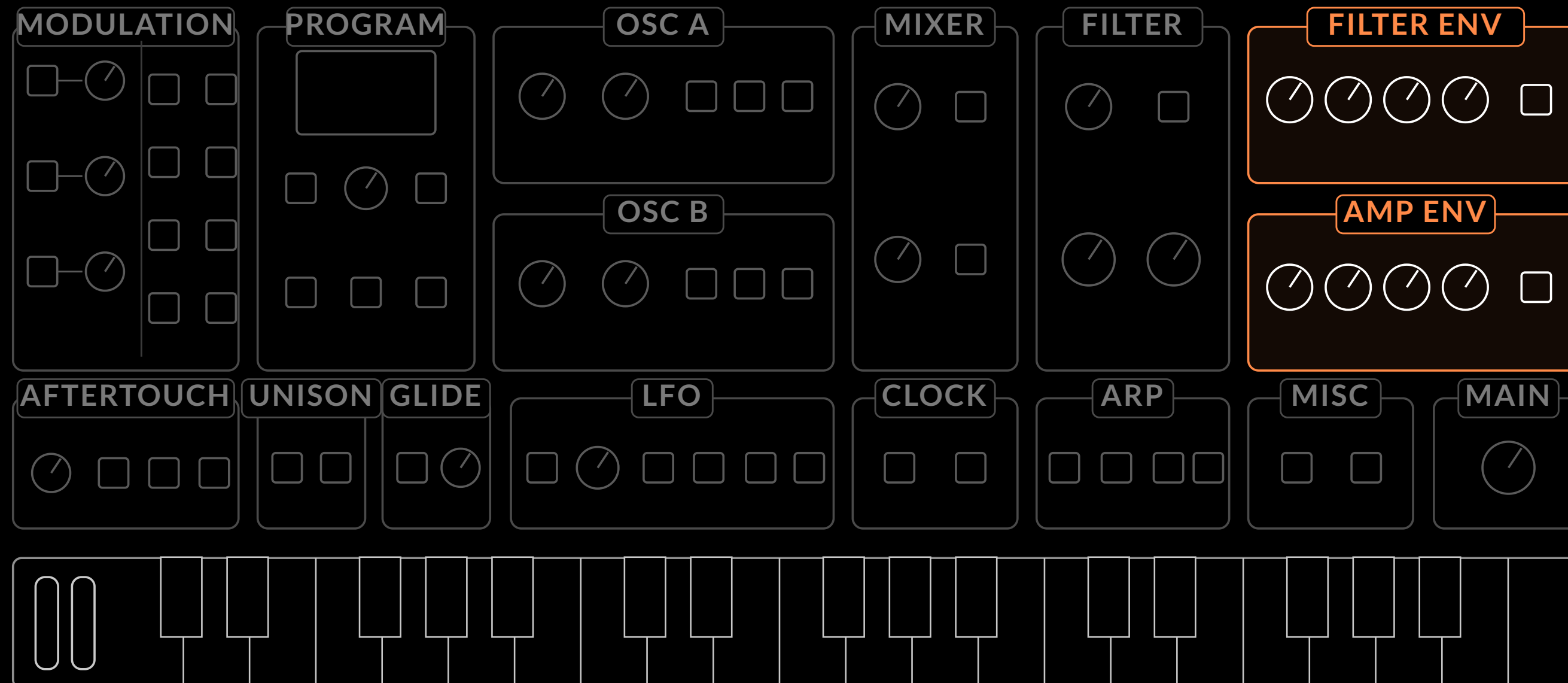
VELOCITY FEEDS BOTH ENVELOPES: FILT ENV VEL & AMP ENV VEL, 0-127

FIVE RESPONSE CURVES IN GLOBAL: SOFT+, SOFT, MEDIUM, HARD, HARD+

WHERE ON FOURM

ENVELOPES › VELOCITY

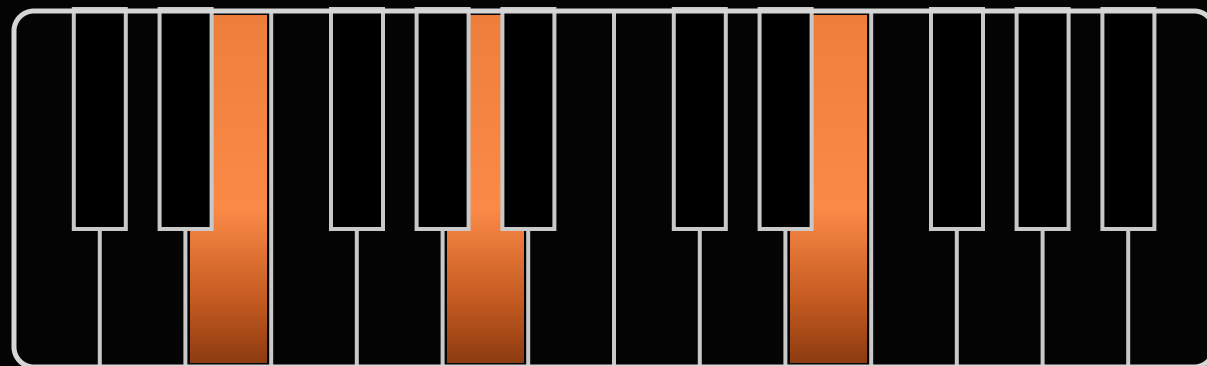
Each envelope has a VELOCITY button: switch it on, and how hard you strike scales that envelope's amount.



POLY AFTERTOUCH

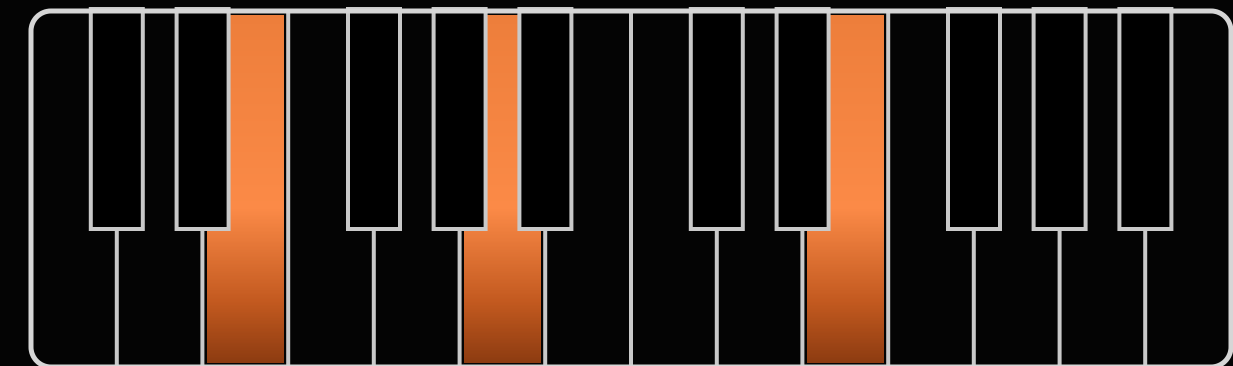
Pressure, per key: the Tactive keybed

MONO AT



ONE PRESSURE VALUE: EVERY VOICE MOVES

POLY AT



A SENSOR UNDER EVERY KEY: ONLY THE PRESSED VOICE MOVES

PRESSURE ROUTING

Press into one note of a chord and only that note opens.

FREQ A

FREQ B

FILT

AMP

LFO FREQ

LFO AMT

PICK DESTINATIONS WITH THE DEST BUTTONS: TWO PER BUTTON

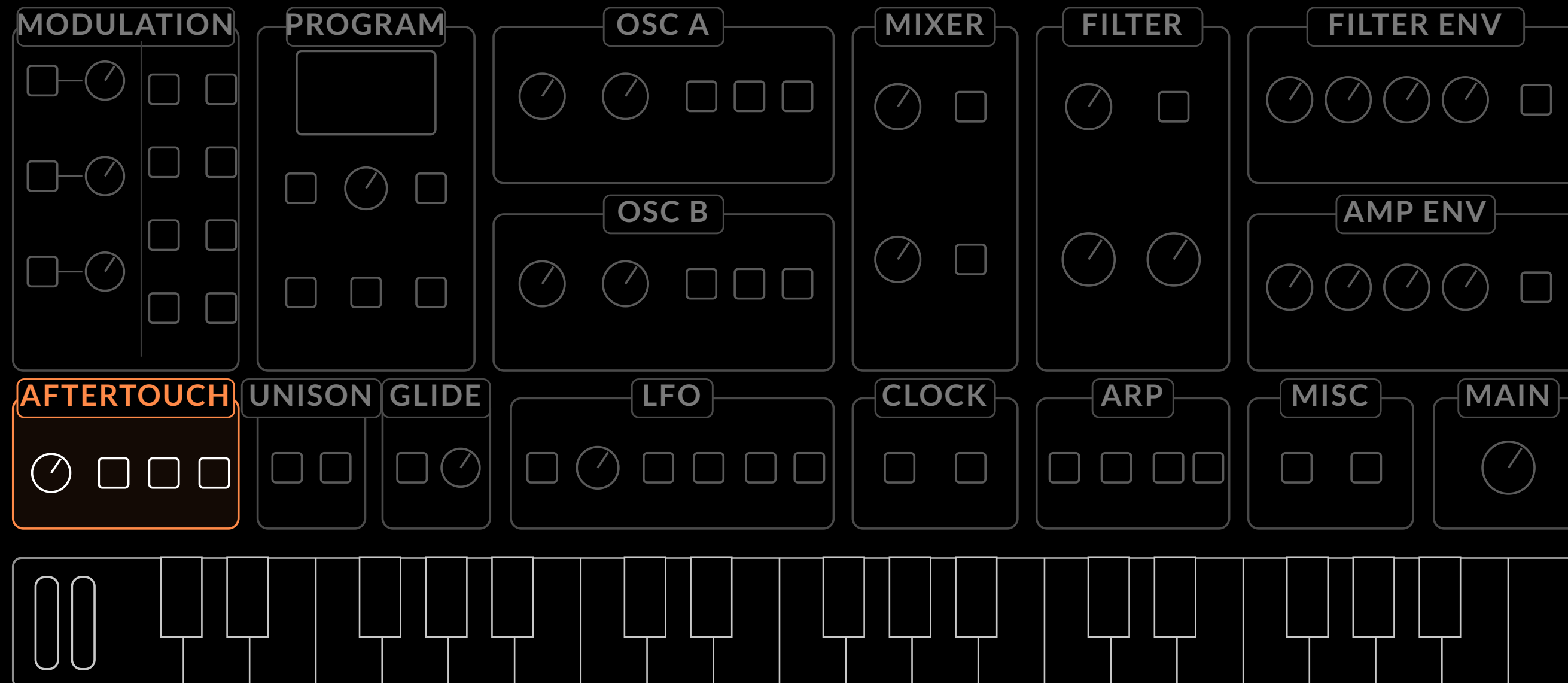
ONE AMOUNT KNOB, POSITIVE OR NEGATIVE

PRESS HARDER WITH +FILT: THE SOUND OPENS · WITH -FILT: IT CLOSES

WHERE ON FOURM

AFTERTOUCH › DESTINATION · AMOUNT

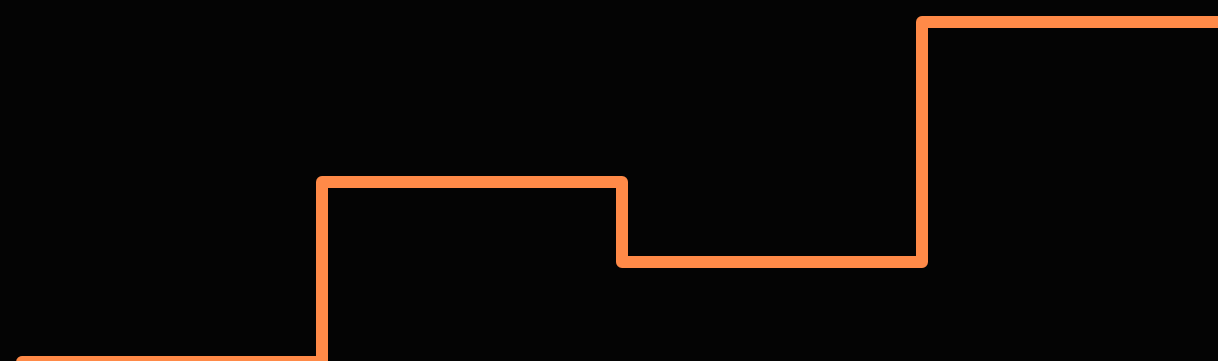
Press harder into a held key. AMOUNT sets how much; the buttons pick where the pressure goes.



GLIDE

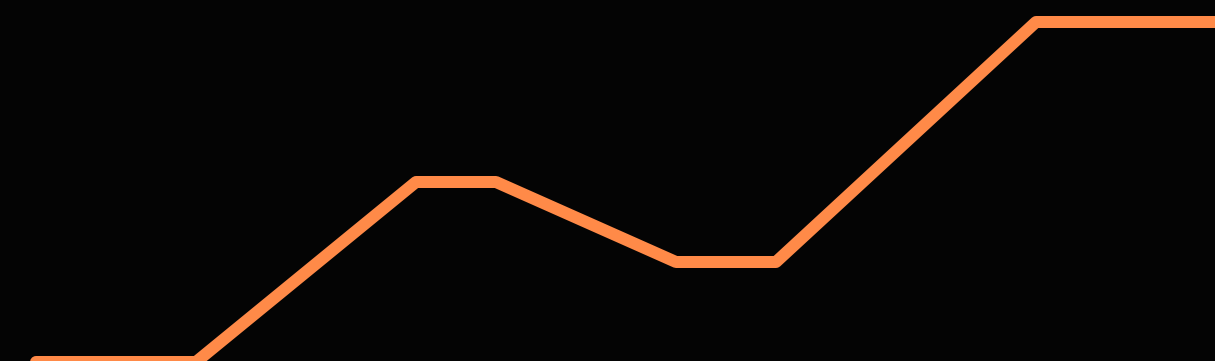
GLIDE on and RATE above zero. Modes: hold GLIDE and turn SELECT.

GLIDE OFF



PITCH JUMPS INSTANTLY

GLIDE ON + RATE



PITCH SLIDES BETWEEN NOTES

GLIDE ON WITH RATE AT 0 DOES NOTHING. SET BOTH

KEY TERMS

New words from this module

VELOCITY

How fast a key goes down. Routed to filter and amp envelope amounts.

POLY vs MONO AT

Per-key pressure vs one value for the whole keyboard.

CURVE

Five velocity curves, from Soft+ to Hard+; aftertouch has its own five.

GLIDE • PORTAMENTO

Pitch slides between notes instead of jumping; legato modes slide only on overlap.

AFTERTOUCH (AT)

Pressure applied after the key bottoms out.

TACTIVE KEYBED

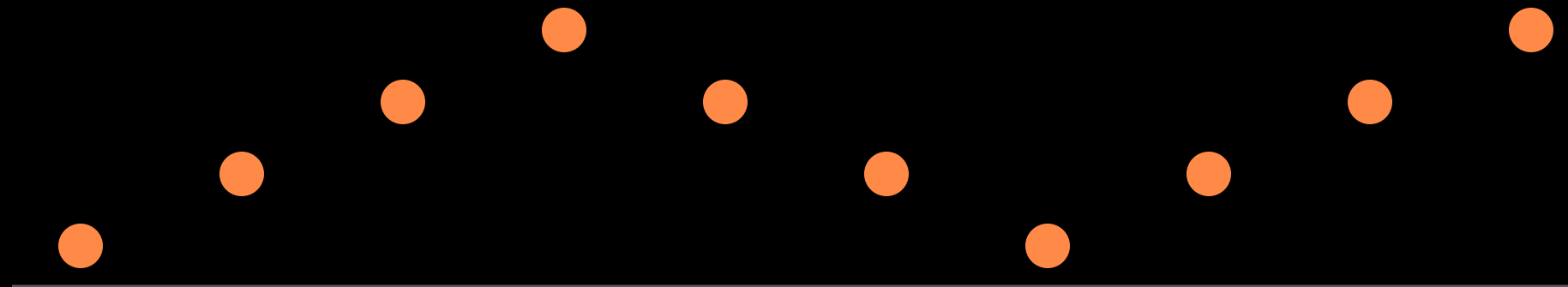
This synth's keybed: a silicone-enclosed pressure sensor per key.

AMOUNT ±

Modulation depth; negative amounts invert the direction.

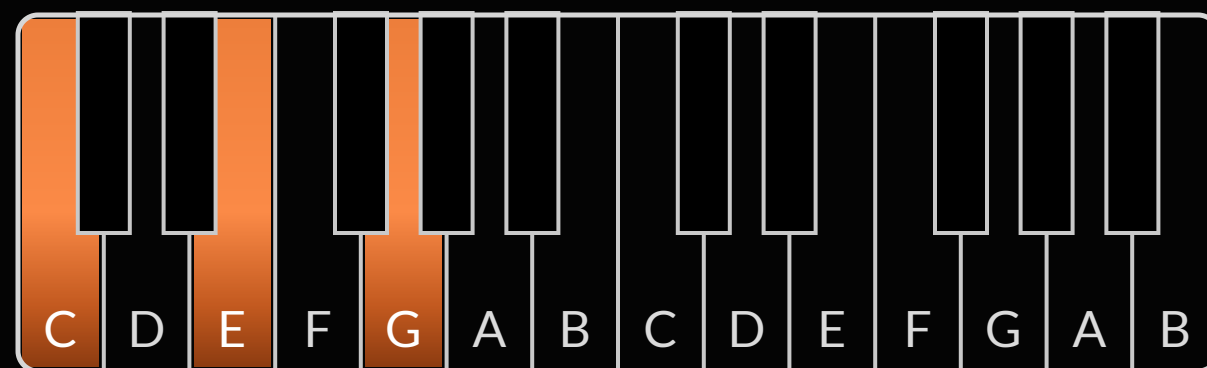
ARPEGGIATOR & SEQUENCER

Patterns from held notes

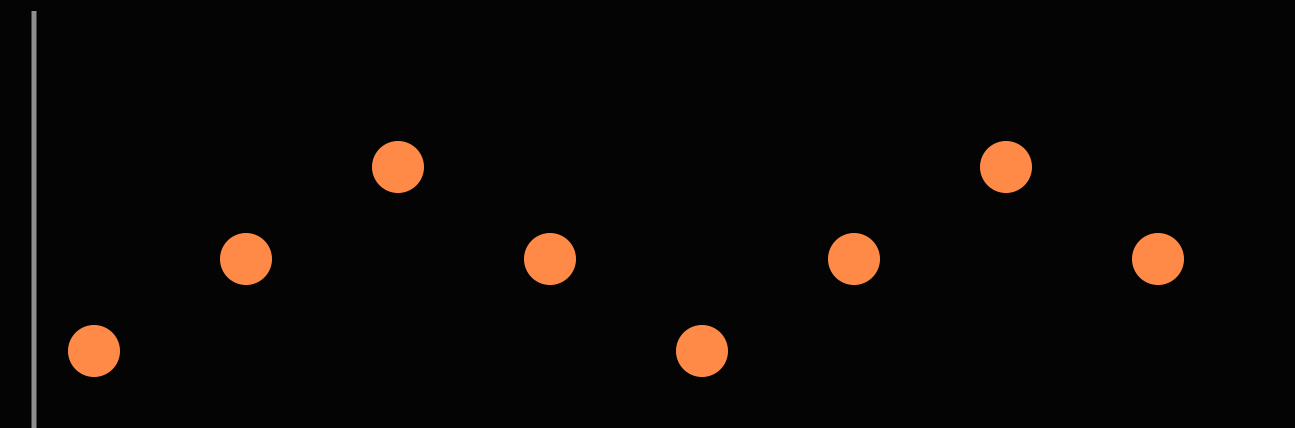


HOLD A CHORD

Hold a chord and the synth plays the pattern.



HOLD C · E · G



UP+DOWN 1 · OCTAVE 1

MODES: UP · DOWN · UP+DOWN 1 & 2 · RANDOM · ASSIGN. PRESS HOLD TO LATCH

WHERE ON FOURM

ARPEGGIATOR › ON · HOLD + CLOCK

ARP on and hold a chord. HOLD: in the UNISON section, latches it after your hands leave; the CLOCK section sets the tempo.



CLOCK & SYNC

Tap TAP TEMPO four times, or hold it and turn SELECT. CLK DIV picks the grid.

Clock Mode set to In: the arp waits for external clock.



SET TEMPO: TAP 4× ON TAP TEMPO · OR HOLD BPM + SELECT

CLK DIV

LFO SYNC

MIDI IN/THRU

WITH MIDI CLOCK MODE IN, THE ARP WAITS FOR EXTERNAL CLOCK

NOTE & MOD SEQ

A sequencer in disguise



RECORD UP TO 64 STEPS: NOTES, RESTS, TIES, EVEN GLIDES

SEQ MOD SENDS STEPS TO: OSC FREQ, CUTOFF, PW, FEEDBACK, MOD AMOUNTS

PLAY ANY KEY: THE WHOLE SEQUENCE TRANSPOSES

KEY TERMS

New words from this module

ARPEGGIATOR

Plays a pattern from the notes of a held chord.

RELATCH

Each new chord starts a fresh arpeggio instead of adding notes.

CLOCK DIVISION

Note value per step: quarters, eighths, sixteenths...

MOD SEQUENCE

Steps that move a parameter instead of playing notes.

HOLD / LATCH

Keeps the arpeggio running after you release the keys.

BPM / TAP TEMPO

The clock speed: dialed in or tapped in.

REST • TIE

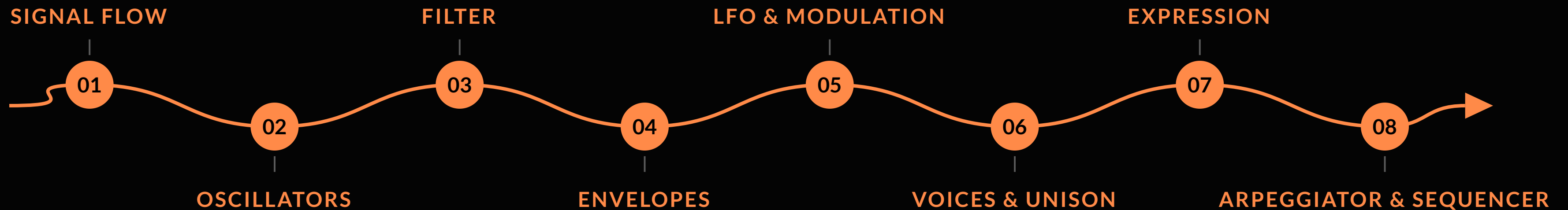
A silent step • a step that extends the previous note.

SEQ REVIEW

Global setting for stepping through and editing a sequence.

THE ROAD, AGAIN

Every synth is this road with different clothes.



THANK YOU

The handbook is yours, EN or RO

SINTEZAUR

the community & what we build next



ZEEDO

try and buy the Fourm



SEQUENTIAL

user guide & presets



SINTEZAUR × ZEEDO · POWERED BY SEQUENTIAL

GLOSSARY

The full glossary is in your handbook

